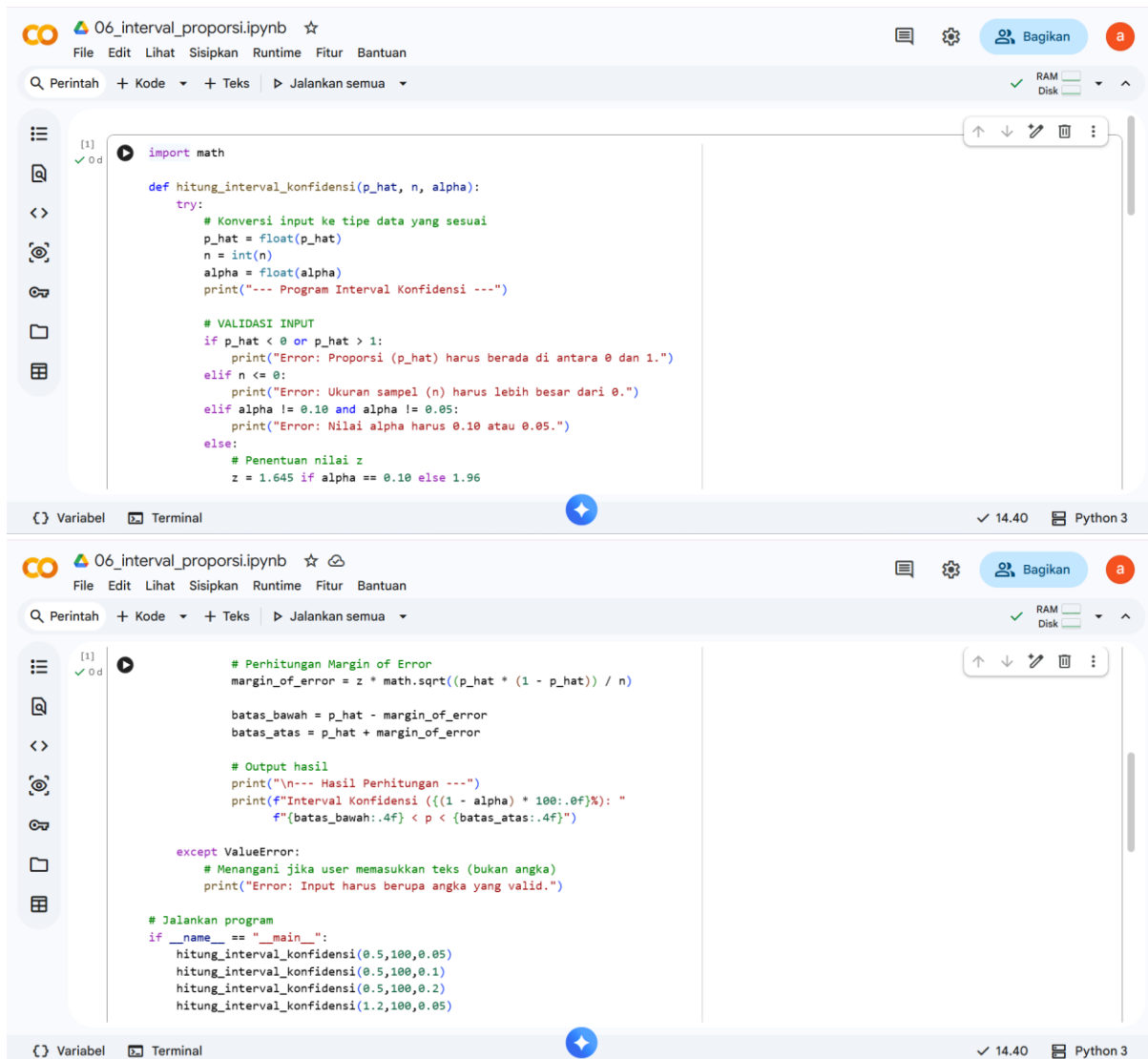


PROGRAM 6

Kodingan :



The image displays two screenshots of a Jupyter Notebook interface, showing the code for a program that calculates a confidence interval for a proportion.

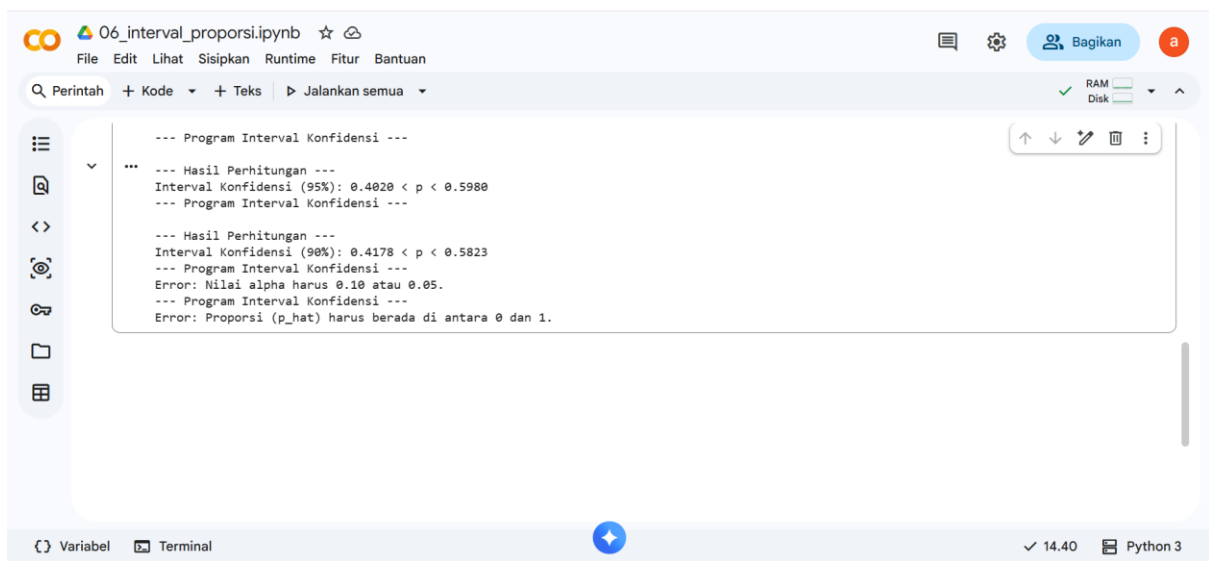
Top Screenshot: The notebook is titled "O6_interval_proporsi.ipynb". The code defines a function `hitung_interval_konfidensi(p_hat, n, alpha)` that takes three arguments: `p_hat` (sample proportion), `n` (sample size), and `alpha` (significance level). The function performs the following steps:

- Converts inputs to appropriate data types: `p_hat = float(p_hat)`, `n = int(n)`, and `alpha = float(alpha)`.
- Prints a header: `print("--- Program Interval Konfidensi ---")`.
- Validates input: Checks if `p_hat` is between 0 and 1, if `n` is greater than 0, and if `alpha` is between 0.10 and 0.05. If any check fails, it prints an error message.
- Determines the z-value: `z = 1.645` if `alpha == 0.10`, otherwise `1.96`.

Bottom Screenshot: This screenshot shows the continuation of the code. It calculates the margin of error and the confidence interval bounds, then prints the results.

- Calculates the margin of error: `margin_of_error = z * math.sqrt((p_hat * (1 - p_hat)) / n)`.
- Calculates the lower and upper bounds: `batas_bawah = p_hat - margin_of_error` and `batas_atas = p_hat + margin_of_error`.
- Prints the results: `print("\n--- Hasil Perhitungan ---")` followed by a formatted string showing the confidence interval: `print(f"Interval Konfidensi ({(1 - alpha) * 100:.0f}%) : " f"{batas_bawah:.4f} < p < {batas_atas:.4f}")`.
- Includes an exception handler for `ValueError` to handle invalid input: `except ValueError: print("Error: Input harus berupa angka yang valid.")`.
- Runs the program with test cases: `hitung_interval_konfidensi(0.5, 100, 0.05)`, `hitung_interval_konfidensi(0.5, 100, 0.1)`, `hitung_interval_konfidensi(0.5, 100, 0.2)`, and `hitung_interval_konfidensi(1.2, 100, 0.05)`.

Hasil *Running* :



The screenshot shows a Jupyter Notebook window titled "O6_interval_proporsi.ipynb". The interface includes a top menu bar with "File", "Edit", "Lihat", "Sisipkan", "Runtime", "Fitur", and "Bantuan". Below the menu is a toolbar with "Perintah", "+ Kode", "+ Teks", and "▶ Jalankan semua". On the right side of the toolbar, there are icons for chat, settings, a "Bagikan" button, and a user profile icon. A status bar at the top right shows "RAM" and "Disk" usage. The left sidebar contains icons for file explorer, search, and other notebook functions. The main area displays the output of a Python program, which is a text-based confidence interval calculator. The output shows two successful calculations for 95% and 90% confidence intervals, followed by two error messages: "Error: Nilai alpha harus 0.10 atau 0.05." and "Error: Proporsi (p_hat) harus berada di antara 0 dan 1." The bottom status bar indicates "Variabel", "Terminal", a blue circular icon, "✓ 14.40", and "Python 3".

```
--- Program Interval Konfidensi ---  
...  
--- Hasil Perhitungan ---  
Interval Konfidensi (95%): 0.4020 < p < 0.5980  
--- Program Interval Konfidensi ---  
  
--- Hasil Perhitungan ---  
Interval Konfidensi (90%): 0.4178 < p < 0.5823  
--- Program Interval Konfidensi ---  
Error: Nilai alpha harus 0.10 atau 0.05.  
--- Program Interval Konfidensi ---  
Error: Proporsi (p_hat) harus berada di antara 0 dan 1.
```